

COMPARATIVE ANALYSIS OF DEEP LEARNING ARCHITECTURES FOR STOCK MARKET DIRECTION PREDICTION USING FINANCIAL NEWS AND MARKET DATA

SYNOPSIS REPORT ON

Deep Learning Project

[ICT - 4442]

Team No. and Names (with Registration Numbers)

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1. Brief Description of the Project

The stock market's movements are not merely the result of previous market data but are also affected by news and investor sentiment. The aim of this project is to compare a number of deep learning models in their ability to predict whether the market will go up or down by using both historical market data and financial news.

The project will make use of DJIA data and daily news headlines from the Daily News for Stock Market Prediction dataset (2008 - 2016), with the news headlines being analysed using FinBERT to extract financial sentiment, after which the sentiment will be combined with market features such as Open, High, Low, Close and Volume.

The task is to carry out a binary classification, aiming to predict the direction of the market for the following trading day. The MLP, CNN, LSTM, and Transformer models will all be trained and compared using accuracy, precision, recall, the F1- score, and confusion matrices.

The project will also investigate models that have been trained with news sentiment and those that have not, in order to find out if financial news has a positive effect on prediction performance. It may also carry out a simple simulated trading analysis using the predictions from the best-performing model.

2. Dataset(s) Identified

Primary Dataset: Daily News for Stock Market Prediction

The project will use the **Daily News for Stock Market Prediction** dataset introduced by Sun (2016). It contains:

- Historical daily news headlines collected from Reddit's WorldNews channel.
- The top 25 headlines for each date.

- News data covering approximately June 2008 to July 2016.
- DJIA market data covering approximately August 2008 to July 2016.
- Approximately 1,989 daily observations in the combined dataset.
- A binary market-movement label associated with the DJIA.

The dataset will be reorganized so that information available on day t is used to predict the market direction on the following trading day, reducing the possibility of temporal information leakage.

3. Proposed Models for Comparison

Model	Architecture Family	Purpose
MLP	Feed-forward neural network	Baseline for learning nonlinear relationships between market and sentiment features
1D CNN	Convolutional neural network	Capture local temporal patterns in market/sentiment sequences
LSTM	Recurrent neural network	Model sequential and longer-term temporal dependencies
Transformer Encoder	Attention-based architecture	Capture relationships between different time steps using self-attention

All four models will receive the same processed input data and will be evaluated on the same temporal test set using common metrics.

FinBERT will be used as a **common pretrained sentiment feature extractor** and will not be counted as one of the four comparative prediction models.

4. Model-to-Member Assignment

Member	Assigned Model / Responsibility
Ashika Mittal	MLP model implementation, experimentation and analysis
Keshav Lakhanpal	1D CNN model implementation, experimentation and analysis
Nishant Bhandari	LSTM model implementation, experimentation and analysis
Rhythm Gaur	Transformer Encoder implementation, experimentation and analysis

5. Existing State-of-the-Art Methods Related to the Proposed Project

Earlier research has shown that both financial news and older market data can help forecast stock market moves.

Ding et al. Used news events, with a CNN while Vargas et al. Looked at CNNs and recurrent models that use news and technical signals.

FinBERT improved financial sentiment analysis more because it was trained only on financial text.

Newer studies also look at Transformers and attention models to capture long-term patterns in data. For example, SA-TrellisNet merges news sentiment with time-based predictions.

This project builds on all that work by giving a comparison of MLP, CNN, LSTM and Transformer models that all use the same data the same features and the same evaluation methods.

6. Objectives

1. To develop a deep-learning-based binary classification system for predicting next-day stock-market direction using historical market information and financial-news sentiment.
2. To compare the predictive performance of MLP, CNN, LSTM and Transformer architectures under a common experimental protocol.
3. To analyse whether incorporating news sentiment improves market-direction prediction compared with using historical market information alone.

7. Importance / Motivation of the Proposed Project

I see that stock markets are shaped by both past market data and outside forces like news and the feelings of investors. Models that depend on price history can overlook crucial details from big news events.

We want to learn how different deep learning designs-MLP, CNN, LSTM and Transformer-handle market and sentiment information. By testing them side by side in the setting we will see which method works best for predicting market moves.

We will also examine whether adding sentiment from news makes predictions better, than using only market data.

8. Preliminary Reference List

[1] X. Ding, Y. Zhang, T. Liu, and J. Duan, “Deep Learning for Event-Driven Stock Prediction,” *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence (IJCAI)*, 2015, pp. 2327-2333.

[2] M. R. Vargas, B. S. L. P. de Lima, and A. G. Evsukoff, “Deep Learning for Stock Market Prediction from Financial News Articles,” *2017 IEEE International Conference on Computational Intelligence and Virtual Environments for Measurement Systems and Applications (CIVEMSA)*, 2017, pp. 60-65.

[3] D. Araci, “FinBERT: Financial Sentiment Analysis with Pre-trained Language Models,” arXiv:1908.10063, 2019.

[4] Z. Zeng, R. Kaur, S. Siddagangappa, S. Rahimi, T. Balch, and M. Veloso, “Financial Time Series Forecasting using CNN and Transformer,” arXiv:2304.04912, 2023.

[5] W.-J. Liu, Y.-B. Ge, and Y.-C. Gu, “News-driven stock market index prediction based on trellis network and sentiment attention mechanism,” *Expert Systems with Applications*, vol. 250, p. 123966, 2024.

9. Work Plan

Sl. No.	Component / Work Element / Milestone	Expected Completion
1	Finalize problem statement, dataset, objectives and model allocation	25/08/2026
2	Complete preliminary literature review and identify relevant papers	31/08/2026
3	Acquire, clean and preprocess market and news data	05/09/2026
4	Implement FinBERT-based sentiment feature extraction	08/09/2026
5	Implement MLP and LSTM baseline models	12/09/2026
6	Implement CNN and obtain preliminary comparative results	18/09/2026
7	Complete Interim report, literature review and contribution log	25/09/2026
8	Implement Transformer model	05/10/2026
9	Perform hyperparameter tuning and final experiments for all models	12/10/2026
10	Conduct model comparison, ablation study and error analysis	18/10/2026
11	Complete final report and presentation	25/10/2026
12	Final code, report and presentation submission	31/10/2026